

APPENDIX · TECHNICAL IMPLEMENTATION

Delivery format?	A web app with phone and desktop layouts from the same page, any browser. Two windows on stage stand in for two phones. Native mobile is roadmap.
Technologies?	Node/Express API, Redis Streams event log, Postgres read models, vanilla JS + WebCrypto AES-GCM, nginx, Docker Compose. Node's built-in test runner. Vended QR library. No frameworks, no CDNs, so the demo needs no internet.
Behind the scenes?	Phone encrypts → POST /events → membership check → XADD to the stream → projector consumer group → Postgres events + handoffs + workload view, then an announcement on a Redis channel that the API relays to every open tab as a server-sent event; a second consumer group, the notifier, writes alerts per subscriber → phones poll and decrypt. The dashboard is computed on the phone.
Built from scratch here?	Spec, requirements and two audit rounds were planned ahead. All code was written during the hackathon window. QR generation is a vended library.
Tradeoffs?	Server-sent events over mobile push, a demo default password over real credential policy, one stream over per-family streams, consent filter deferred. Each one on the honesty slide with the reason.

APPENDIX · IMPACT AND USERS

Target user?	Adult children sharing care of an elderly parent. The elder is the subject and owner of the preferences, not the app user. Support workers join with their own role.
Discovery and adoption?	Family to family, by QR, in person, by design. Community care programs and masjid welfare committees as the channel. Agencies bring their families.
How many benefit?	1 in 4 American Muslims already care for an older adult (ISPU 2026). Muslims are 8.7% of Greater Montreal (Census 2021). Start with Montreal families, then any family with a shared care load.
Impact if fully developed?	Fewer missed meds and appointments, a fairer split of the load, and less of the strain ISPU documents. A family-owned health record the family never had.
Success measure?	Plan adherence, handoff acceptance time, active caregivers per family, weekly return rate.

APPENDIX · DEMO, FUNCTIONALITY, BUSINESS

Fully functional or mocked?	The core loop is real end to end: encrypt, stream, project, store, decrypt. Login is real, with a demo default password. Updates are live over server-sent events, with polling as the safety net. The consent filter is not built. Nothing on stage is faked.
Most important feature?	The record itself, private by construction. From it: the handoff that writes itself, and the hospital sheet the ER can read in a minute.
Hardest part?	Keeping the server useful while it can read nothing: deciding which fields stay clear for routing and counting, and building a whole dashboard from client-side decryption.
Business model?	Free for families. Paid per elder per month by home-care agencies and community programs that need handoff logs and scoped access.
Scale beyond the prototype?	Per-family streams, push, scoped keys, password policy, offline. Ciphertext-only storage keeps compliance and hosting light.
Who pays?	Agencies and community organisations. Families optionally for extras like key escrow between two relatives.
What would it take to make it a real product?	Password policy and key recovery, native apps, scoped support keys, push, offline, accessibility, Urdu and Arabic.

APPENDIX · TEAM, PROCESS, DATA AND PRIVACY

How did you divide the work?	Abdul: architecture, event model, crypto, api and projector, dashboard, tests. safio: in-app invite flow, interface cleanup, the flow view that traces a write hop by hop.
What did you learn?	Cut scope early and say so. One workflow done well beats five half done. Privacy shapes every feature, including which ones you cannot build yet.
Differently next time?	Tests from hour one. Push notifications budgeted from the start.
How did you prioritise?	Requirement register of 46 items, two audit rounds on feasibility in 24 hours, then "cut core": the handoff loop first, everything else labelled roadmap.
What data?	Care events, handoff summaries, preferences, routine, member names, all encrypted. Clear: event type, category, member id, timestamps.
Ethical risks?	Support workers currently hold the same key, so hiding categories is UI only. The password-locked key copy is only as strong as the password, and the demo default is 333. Lost key means lost data until rotation ships. No diagnosis, dosage or triage anywhere, and the metrics are counts, never judgments.
Depends on an LLM?	No. By design the server cannot run one on this data.

APPENDIX · FUTURE AND CLOSING

Next if more time?	Urdu and Arabic for the elder view, scoped support key, push, key rotation, password policy and recovery.
Continue after today?	Yes. Five-family pilot in Montreal next week.
First step after today?	Commit and publish the repo, deploy on one small VM, recruit the pilot families through community care contacts.
Different from existing options?	Task apps organise chores, med apps remind, marketplaces find carers. None connect plan, actuals, handoff and load with the elder's preferences, and none let only the family read the data.
Help needed?	Introductions to community care programs and agencies, pilot families, and a designer for the elder view.
Remember most?	The server cannot read it. Tap Server view on any screen, or open the database, and see for yourself.
Why should this win?	It solves the coordination problem the track asked for, with the elder's preferences built in, privacy that is cryptographic rather than a promise, an honest scope, and a test suite behind it.

APPENDIX • SOURCES

- ISPU, *Support and Sacrifice: Muslims Who Care for Older Adults Face Emotional Strain*, 2026, drawing on American Muslim Poll 2025. ispu.org/reports-and-analysis/support-and-sacrifice/
- ISPU, *Sandwiched: Muslims Are Most Likely to Be Caring for Aging Parents and Have School-Aged Children*. ispu.org/muslim-families/
- Statistics Canada, *More than half of women provide care to children and care-dependent adults in Canada, 2022*, The Daily, 8 Nov 2022. www150.statcan.gc.ca/n1/daily-quotidien/221108/dq221108b-eng.htm
- Statistics Canada, *Study: "Sandwiched" between multiple unpaid caregiving responsibilities*, The Daily, 2 Apr 2024. www150.statcan.gc.ca/n1/daily-quotidien/240402/dq240402d-eng.htm
- Collins, R. N. and Kishita, N., *Prevalence of depression and burden among informal care-givers of people with dementia: a meta-analysis*, Ageing & Society, 2020.
- MuslimHacks 2026 Challenge Guide, Elderly Care (research and fact sheets by Process Plus). References EC1 to EC6 therein, including Law 25, PHIPA, PIPEDA and Attum et al., *Cultural Competence in the Care of Muslim Patients and Their Families*, StatPearls.
- Statistics Canada, Census 2021 religion release: Muslims 4.9% of Canada, 8.7% of Greater Montreal (cited in the Challenge Guide).
- Chrome for Developers, *Built-in AI: Prompt, Summarizer and Translator APIs on Gemini Nano*. developer.chrome.com/docs/ai/built-in